Non-Transitive Chord: Bounded Multi-Predecessor Stabilization and Gap-Bridge Delegation over Restricted WebRTC Graphs

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***Abstract*—Distributed hash tables such as Chord assume a transitive underlay: if node a can communicate with b and b can communicate with c, then a can open a direct transport channel to c once b advertises c's address. In browser and native WebRTC overlays, symmetric NATs, enterprise firewalls, and disjoint private LANs routinely violate transitivity, yielding connected underlay graphs composed of full-mesh subcliques joined at one or more articulation (bridge) nodes. When standard Chord stabilization and greedy delivery execute over such a graph, three structural failures arise: (1) stabilization reports whose newly advertised peers fail ICE negotiation are discarded when confirmation filters leave the report empty, stalling predecessor notifications while re-dialing the same unreachable candidates on every tick; (2) a single predecessor slot at an articulation node is preempted by the closer sub-clique's predecessor, starving the farther sub-clique's local finger proof and connection retention; and (3) greedy clockwise routing across interleaved identifier rings enters a local minimum that single-crossing handoff cannot escape without knowing which linked neighbor bridges the missing ring arc.**

Prior work addresses non-transitive DHTs via unstructured neighborhood gossip or global virtual links in the finger table, but both mechanisms violate the $O(1)$ control-frame and slot-accounting invariants required of a browser-safe WebRTC overlay. We present a protocol extension that resolves all three failure modes with zero extra control messages, $O(K)$ report payloads, and a strictly direct-only finger table. Articulation nodes maintain a counter-clockwise ordered sequence of up to K direct predecessors; stabilization and successor-sync handlers extract passive 1-hop gap-bridge hints from the unconfirmed entries of already-authenticated `TopoInfo` reports; failed candidate handshakes enter a bounded negative-cache cooldown; and delivery delegates across unroutable ring gaps in a single hop by reusing the existing RouteStage $=$ aim $\times\mathbb{B}$ **carrier. We prove preservation of all route-safety laws (`Linked`, `NeverOvershoots`, `SingleCrossing`, `Acyclic`, and the $2|V|$ hop bound) and establish all-pairs delivery in at most 3 hops across bridge-connected cliques.**

***Index Terms*—Chord, Distributed Hash Tables, Non-Transitive Connectivity, WebRTC, NAT Traversal, Stabilization, Formal Verification.**

I. Introduction and Failure Analysis

The Chord lookup protocol [1], [2] and its formally verified descendants [3] organize nodes on the identifier ring $\mathbb{Z}_{2^{160}} = \mathbb{Z}/2^{160}\mathbb{Z}$ under the clockwise distance metric

$$\mathrm{d}(a, b) = (b - a) \bmod 2^{160}. \tag{1}$$

Correctness of the classical stabilization loop rests on an implicit underlay assumption: whenever $b = \mathrm{head}(a)$ reports a predecessor or successor $p \in (a, b)$ that is closer to a on $\mathbb{Z}_{2^{160}}$, a can establish a direct link to p and adopt p as its new successor head.

In real-world deployments [4], [5], physical reachability is governed by an undirected graph $G = (V, L)$ that is connected but not transitive ($L \subsetneq \binom{V}{2}$). Consider the canonical two-clique topology joined at a single dual-homed or public bridge node C:

$$V_1 = \{A, B, C\}, \quad V_2 = \{C, D, E\}, \quad L = \binom{V_1}{2} \cup \binom{V_2}{2}, \tag{2}$$

where nodes in $V_1 \setminus \{C\}$ and $V_2 \setminus \{C\}$ cannot complete ICE/DTLS handshakes with one another. Because node identifiers are cryptographic hashes of public keys, their cyclic order on $\mathbb{Z}_{2^{160}}$ interleaves the two cliques. Without loss of generality, consider the interleaved ring order

$$A(10) \prec B(30) \prec E(40) \prec C(50) \prec D(80) \prec A(10). \tag{3}$$

Figure 1 illustrates this configuration. In V_1, the sub-ring successor of $B(30)$ is $C(50)$; in V_2, the sub-ring successor of $E(40)$ is also $C(50)$, while the sub-ring successor of $D(80)$ is $E(40)$. Tracing standard stabilization and routing over (3) exposes three coupled defects:

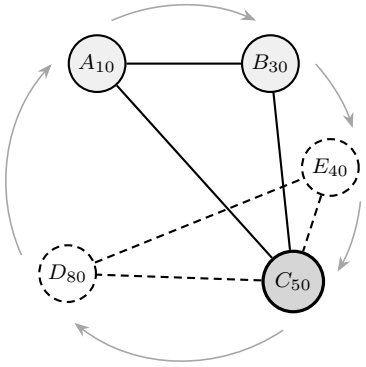


Fig. 1. Interleaved ring $A(10) \to B(30) \to E(40) \to C(50) \to D(80)$ over two non-transitive WebRTC cliques $V_1 = \{A, B, C\}$ (solid links) and $V_2 = \{C, D, E\}$ (dashed links) sharing bridge node $C(50)$.

Formal specification and verification note for `rings-core` (`crates/core/src/dht/topology`, `crates/core/src/dht/delivery`, and `crates/core/src/swarm/transport`), issue #1195.

A. Failure Mode 1: Empty-Confirmed Report Drop and Dial Amplification

When a node u receives a stabilization report `reported` from its current head $r = \mathrm{head}(u)$, it dials bounded unknown candidates and then filters the report under the transport lifecycle lock to retain only routable peers:

$$\texttt{confirmed} = \texttt{reported}.\,\mathrm{confirmed_by}\big(p \mapsto p = u \vee \mathrm{Routable}(u,p)\big).$$

Suppose $r = C(50)$ reports only peers in $V_2 \setminus \{C\} = \{D, E\}$ to a node $u \in V_1 \setminus \{C\}$ (for instance before C has populated A or B into its successor list, or when K slots at C are occupied by nodes of V_2). Because u cannot complete ICE with D or E, `confirmed` is empty ($\neg$`confirmed`.has_confirmed_peer()). Guarding the commit on `confirmed`.has_confirmed_peer() aborts the stabilization transition entirely: u never commits $\texttt{Stabilize}(r)$, never emits $\texttt{Notify}(r)$ to inform C of u's existence, and on the next periodic tick after pending handshake expiry, u re-dials D and E from scratch, generating an unbounded stream of relayed `ConnectNodeSend` SDP offers across C.

B. Failure Mode 2: Single-Predecessor Preemption at Articulation Nodes

In (3), both $B(30) \in V_1$ and $E(40) \in V_2$ have $\mathrm{head}(B) = \mathrm{head}(E) = C(50)$ within their respective reachable subcliques, so both send $\texttt{Notify}(C)$. Under a single predecessor register `predecessor` : Option⟨Did⟩, the update rule

$$\mathrm{rectify}(n, p_{\mathrm{cur}}, p_{\mathrm{cand}}) : \mathrm{d}(n, p_{\mathrm{cand}}) > \mathrm{d}(n, p_{\mathrm{cur}})$$

retains only $E(40)$ at $C(50)$ because $\mathrm{d}(C(50), E(40)) = 246 > \mathrm{d}(C(50), B(30)) = 236$. Whenever $B(30)$ notifies $C(50)$ first and $E(40)$ notifies second, $E(40)$ overwrites $B(30)$; whenever $E(40)$ notifies first and $B(30)$ notifies second, $B(30)$ is ignored. Consequently:

1) When $C(50)$ answers $B(30)$'s stabilization query, $C(50)$ reports only `predecessor` = Some($E(40)$), which $B(30)$ strips as unroutable. Thus $B(30)$ observes `confirmed.predecessor` = None $\neq$ Some($B(30)$) and can never prove its local finger range $[0..\ell(B, C)]$ [3].
2) If $|V_2 \setminus \{C\}| \geq K$, C's capacity-K successor list is filled by nodes of V_2, so $B(30)$ is absent from both C's successor list and C's single predecessor slot, leaving $B(30) \notin \mathrm{Referenced}(C)$ and exposing the critical bridge link (B, C) to connection-capacity eviction.

C. Failure Mode 3: Routing Dead-End and False Local Termination

Consider a payload originated by $E(40)$ or $D(80)$ in V_2 toward $A(10)$ or $B(30)$ in V_1:

1) At $E(40)$, the linked known peers are $\{C(50), D(80)\}$. For target $A(10)$, $\mathrm{d}(E(40), A(10)) = 226$, while $\mathrm{d}(E(40), C(50)) = 10$ and $\mathrm{d}(E(40), D(80)) = 40$. Both $C(50)$ and $D(80)$ lie in the half-open arc $(E(40), A(10)]$. Greedy clockwise forwarding selects $\arg\max_{p \in (E,A]} \mathrm{d}(E, p) = D(80)$, sending the packet *away* from the bridge $C(50)$ to $D(80)$.
2) At $D(80)$, the linked known peers are $\{E(40), C(50)\}$ at distances $\mathrm{d}(D(80), E(40)) = 216$ and $\mathrm{d}(D(80), C(50)) = 226$, both strictly greater than $\mathrm{d}(D(80), A(10)) = 186$. Thus no linked peer of $D(80)$ lies in $(D(80), A(10)]$.
3) Standard single-crossing handoff at $D(80)$ forwards to the nearest clockwise linked peer $\arg\min_p \mathrm{d}(D(80), p) = E(40)$ with $\texttt{handed_off} = \top$.
4) Upon returning to $E(40)$ with $\texttt{handed_off} = \top$, greedy forwarding again selects $D(80)$, where a second crossing is refused and the packet is dropped ($\perp$).

Thus the entire trace $E(40) \to D(80) \xrightarrow{\text{handoff}} E(40) \to D(80) \to \perp$ bypasses bridge $C(50)$ and fails to deliver, even though a 2-hop physical path $E(40) \to C(50) \to A(10)$ (and $D(80) \to C(50) \to A(10)$) exists in G.

D. Why Unstructured Gossip and Virtual Fingers Are Inadmissible

Two classical workarounds from the peer-to-peer literature are incompatible with the resource and safety invariants of `rings-core`:

- **Unstructured 1-hop neighborhood gossip** (broadcasting all active connections in `TopoInfo` and dialing every 2-hop neighbor) inflates `TopoInfo` from $O(K)$ to $O(\deg(n))$ and triggers $O(N^2)$ WebRTC SDP offer storms across bridge nodes, violating the $O(1)$ control-message bound and per-slot memory accounting.
- **Global virtual links in `FingerTable`** [6], [7] (inserting unlinked peers into finger slots and tunneling every Chord hop over multi-hop overlays) break the fundamental transport-topology invariant $\mathrm{Referenced}(n) \subseteq \mathrm{Admitted}(n)$ and can amplify a single $O(\log N)$ lookup into $O(\log^2 N)$ physical hops.

II. System Model and State Carrier

Let $V \subseteq \mathbb{Z}_{2^{160}}$ be the set of live nodes and $L \subseteq \binom{V}{2}$ be the set of active, routable WebRTC data-channel links. For each node $n \in V$, let $\mathrm{Admitted}(n) = \{p \in V \setminus \{n\} : \{n, p\} \in L\}$ be its set of admitted transport peers, and let $K \geq 1$ denote the successor and predecessor capacity (`DEFAULT_SUCCESSOR_CAPACITY` = 3).

Definition 1 (Extended Topology Carrier)**.** *The local topology state of node $n \in V$ is the tuple*

$$S_n = \big(n,\ \mathit{succ}_n,\ \mathit{pred}_n,\ \mathit{preds}_n,\ \mathit{fingers}_n,\ \mathit{bridged}_n,\ \Phi_n,\ \Pi_n\big), \quad (4)$$

where:

- $\mathit{succ}_n \in (V \setminus \{n\})^{\leq K}$ *is strictly ordered by ascending clockwise distance* $\mathrm{d}(n, \cdot)$*;*

- $\texttt{preds}_n \in (V \setminus \{n\})^{\le K}$ *is strictly ordered by* descending *clockwise distance* $\mathrm{d}(n,\cdot)$ *(i.e., ascending counter-clockwise distance* $\mathrm{d}(\cdot,n)$*);*
- $\texttt{pred}_n = \texttt{preds}_n.\mathrm{first}() \in \mathrm{Option}\langle\mathrm{Did}\rangle$ *is the primary predecessor projection;*
- $\texttt{fingers}_n \in (\mathrm{Option}\langle\mathrm{Did}\rangle)^{160}$ *is the sparse/no-wrap direct finger table;*
- $\texttt{bridged}_n : \mathrm{Did} \rightharpoonup \mathrm{Did}$ *is a finite partial map from unconfirmed remote peer* p *to an admitted bridge reporter* $b = \texttt{bridged}_n(p)$*;*
- Φ_n *and* Π_n *are the finger-convergence and pending-stabilization protocol states.*

Definition 2 (Referenced Peers and Direct-Link Invariant)**.** *The set of topology-referenced peers at node* n *is*

$$\mathrm{Referenced}(S_n) = \big(\mathrm{set}(\texttt{succ}_n) \cup \mathrm{set}(\texttt{preds}_n) \cup \{p \mid \exists i.\ \texttt{fingers}_n[i] = \mathrm{Some}(p)\}\big) \setminus \{n\}. \quad (5)$$

The state S_n *satisfies* `TopologyReferencesOnlyAdmitted` *iff*

$$\mathrm{Referenced}(S_n) \subseteq \mathrm{Admitted}(n), \quad (6)$$

and every bridge witness satisfies

$$\forall (p \mapsto b) \in \texttt{bridged}_n : \\ p \neq n \ \wedge\ b \in \mathrm{Referenced}(S_n) \ \wedge\ p \notin \mathrm{Referenced}(S_n). \quad (7)$$

Equations (6) and (7) preserve a strict separation of concerns: $\mathrm{Referenced}(S_n)$ contains *only* directly admitted WebRTC peers (protecting all of $\texttt{succ}_n$, $\texttt{preds}_n$, and $\texttt{fingers}_n$ from unreferenced-connection eviction), while $\texttt{bridged}_n$ stores passive 1-hop reachability hints whose 'via' peer b is always an admitted member of $\mathrm{Referenced}(S_n)$.

III. Bounded Multi-Predecessor Stabilization

A. Empty-Confirmed Commit and Candidate Dial Backoff

When node n processes an authenticated stabilization report `reported` from r echoing the active token `request_id`, n partitions the advertised peers under the lifecycle lock using the routability predicate $\rho_n(p) \equiv (p = n) \vee \mathrm{Routable}(n,p)$:

$$\texttt{confirmed} = \texttt{reported}.\mathrm{confirmed_by}(\rho_n), \quad (8)$$

$$\texttt{unconfirmed} = \texttt{reported}.\mathrm{unconfirmed_peers}(n,\rho_n). \quad (9)$$

So long as the reporter itself remains an active routable connection ($\mathrm{Routable}(n,r)$), n commits the stabilization transition even when `confirmed` contains no peers:

$$\mathrm{Routable}(n,r) \implies \mathrm{commit}\big(\texttt{Stabilize}(r, \texttt{request_id}, \texttt{confirmed}, \texttt{unconfirmed})\big). \quad (10)$$

Because $r = \mathrm{head}(n) \in \texttt{succ}_n$ is already confirmed by $\mathrm{Routable}(n,r)$, $\mathrm{head}(\texttt{next_succ}_n)$ remains non-empty and emits $\texttt{Notify}(r)$, breaking the notification deadlock of Failure Mode 1.

To prevent repeated SDP offers toward unreachable candidates across periodic stabilization ticks, each transport maintains a bounded negative-cache $\mathcal{B}_n : \mathrm{Did} \rightharpoonup (\mathrm{failures} \in \mathbb{N}^+, \mathrm{retry_after_ms} \in \mathbb{N})$ of capacity $B_{\max} = 64$. Whenever an unadmitted handshake attempt for candidate c fails or expires before admission, $\mathcal{B}_n$ records a cooldown

$$\Delta_{\mathrm{backoff}}(f) = \min\big(\Delta_0 \cdot 2^{\min(f-1,4)},\ \Delta_{\max}\big), \quad (11)$$

with $\Delta_0 = 30\,\mathrm{s}$ and $\Delta_{\max} = 300\,\mathrm{s}$. Automatic DHT candidate dials (`ConnectDhtPeer`) skip c while $\mathrm{now_ms} < \mathrm{retry_after_ms}(c)$, whereas explicit application connects (`Swarm::connect`) bypass and clear $\mathcal{B}_n(c)$, and successful admission of c immediately clears $\mathcal{B}_n(c)$.

B. Multi-Predecessor Rectification Law

Let $\mathrm{can_rectify}(S_n, c)$ be the geometric admissibility predicate requiring $c \neq n$ and $c \notin (n, \mathrm{head}(S_n))$ (with $c = \mathrm{head}(S_n)$ allowed only on a two-node wrapping ring). We define the bounded multi-predecessor rectification operator $\mathrm{rectify_preds}(S_n, c, K)$ as follows:

Definition 3 (Supersede-on-Closer, Retain-on-Sub-Ring)**.** *Given state* S_n*, candidate* c*, and capacity* $K \ge 1$*:*

1) *If* $\neg\,\mathrm{can_rectify}(S_n, c)$ *or* $K = 0$*, return* $\texttt{preds}_n$*.*
2) *Let* $p_0 = \texttt{preds}_n.\mathrm{first}()$*.*
3) ***(Supersede on closer joiner)*** *If* $p_0 = \mathrm{None}$ *or* $\mathrm{d}(n,c) > \mathrm{d}(n,p_0)$*, then* c *lies strictly between* p_0 *and* n *on* $\mathbb{Z}_{2^{160}}$*; return the singleton* $[c]$ *(any sub-ring predecessor behind* c *that cannot reach* c *will re-notify* n *on its next stabilization tick).*
4) ***(Retain on sub-ring notification)*** *If* $\mathrm{d}(n,c) \le \mathrm{d}(n,p_0)$*, insert* c *into* $\texttt{preds}_n$*, sort by descending* $\mathrm{d}(n,\cdot)$*, deduplicate, and truncate to* K*.*

Lemma 1 (Transitive Fixpoint Collapse and Articulation Retention)**.** *Under Definition 3:*

1) *On any transitive ring* $(V, \binom{V}{2})$ *after stabilization converges, only the unique global predecessor* $p^* = \mathrm{Pred}_V(n)$ *has* $\mathrm{head}(p^*) = n$*; hence* $\texttt{preds}_n = [p^*]$ *of length* 1*.*
2) *On a non-transitive graph* $G = (V, L)$ *where* $m \le K$ *cliques* $V_1, \ldots, V_m$ *meet at articulation node* C*, let* $p_i = \mathrm{Pred}_{V_i}(C)$ *be* C*'s predecessor in each clique* V_i*. Because* $\mathrm{head}_{V_i}(p_i) = C$*, each* p_i *periodically commits stabilization against* C *and emits* $\texttt{Notify}(C)$*. Within at most two stabilization rounds after the last join,* $\texttt{preds}_C$ *retains all* m *predecessors ordered by descending* $\mathrm{d}(C,p_i)$*, and* $\{p_1, \ldots, p_m\} \subseteq \mathrm{Referenced}(S_C)$*.*

Proof. Part (1) follows because any earlier predecessor q with $\mathrm{d}(n,q) < \mathrm{d}(n,p^*)$ is superseded when p^* notifies n, and once q learns p^* from n's `TopoInfo`, transitivity allows q to connect to p^* and move its head into $(q, p^*]$, so q never notifies n again. For Part (2), let $p_{(1)}, \ldots, p_{(m)}$ be ordered so that $\mathrm{d}(C,p_{(1)}) > \cdots > \mathrm{d}(C,p_{(m)})$. Once $p_{(1)}$ notifies C, $\texttt{preds}_C[0] = p_{(1)}$ can never be superseded by

any $p_{(j)}$ $(j > 1)$. On the subsequent stabilization tick of each $p_{(j)}$ $(j > 1)$, (10) commits $p_{(j)}$'s report from C and emits $\texttt{Notify}(C)$, which satisfies Clause 4 of Definition 3 and inserts $p_{(j)}$ into $\texttt{preds}_C$ up to capacity $K \geq m$. Since $\text{set}(\texttt{preds}_C) \subseteq \text{Referenced}(S_C)$, connection eviction never retires any $p_{(j)}$. □

C. Local Finger Range Proof under Multi-Predecessor Reports

Each `TopoInfo` frame serialized by reporter r carries both $\texttt{successors} = \texttt{succ}_r$ and $\texttt{predecessors} = \texttt{preds}_r$ (at most $2K$ DIDs in total). When receiver n filters r's report via confirmed_by(ρ_n),

$$\texttt{confirmed.preds} = [p \in \texttt{reported.preds} \mid \rho_n(p)], \quad (12)$$

$$\texttt{confirmed.pred} = \texttt{confirmed.preds}.\,\text{first}(). \quad (13)$$

Lemma 2 (Sub-Clique Local Finger Proof)**.** *In the two-clique ring* (3)*, once* $C(50)$ *holds* $\texttt{preds}_C = [E(40), B(30)]$*,* $B(30)$*'s confirmation filter over* C*'s report yields* $\texttt{confirmed.preds} = [B(30)]$ *and* $\texttt{confirmed.pred} = \text{Some}(B(30))$*. Consequently,* `stabilized_successor_proof_end` *at* $B(30)$ *proves all finger slots* i *with* $2^i \leq \text{d}(B(30), C(50))$ *to* $C(50)$*.*

Proof. At $B(30)$, $\rho_B(E(40)) = \bot$ and $\rho_B(B(30)) = \top$. Thus the closest routable predecessor of $C(50)$ in $B(30)$'s clique V_1 is $B(30)$ itself, so $\texttt{confirmed.pred} = \text{Some}(B(30))$. Since $\text{head}(S_B) = \text{head}(\texttt{next_succ}_B) = C(50)$ and no peer in Admitted(B) lies in $(B(30), C(50))$, $C(50)$ is the exact direct finger for every slot threshold $2^i \leq \text{d}(B(30), C(50))$. □

IV. Passive Gap-Bridge Hints and Owner Lookup

A. Zero-TTL Passive Hint Lifecycle

Whenever node n processes an authenticated `TopoInfo` report from an admitted successor $r \in \texttt{succ}_n$ (via either `QueryFor::Stabilization` or `QueryFor::SyncSuccessor`), the unconfirmed peers

$$U(n, r) = \big(\text{set}(\texttt{reported.preds}) \cup \text{set}(\texttt{reported.succ})\big) \setminus \{p \mid \rho_n(p)\}$$

are peers to which r holds an active link ($\{r, p\} \in L$) while n does not ($\{n, p\} \notin L$). The pure transition updates $\texttt{bridged}_n$ according to three deterministic rules:

1) **Report refresh (`Stabilize` / `SyncSuccessor`):** Remove all prior entries $(p \mapsto b) \in \texttt{bridged}_n$ with $b = r$, then for each $p \in U(n, r)$ with $p \notin \text{Referenced}(S'_n)$, insert $p \mapsto r$.
2) **Direct admission ($\texttt{Admit}(p)$):** Remove p from $\text{dom}(\texttt{bridged}_n)$ because p is now directly linked.
3) **Peer removal ($\texttt{Remove}(p)$):** Retain only entries $(t \mapsto b) \in \texttt{bridged}_n$ with $t \neq p \wedge b \neq p$.

Lemma 3 (Boundedness and Liveness of $\texttt{bridged}_n$)**.** *For every reachable state* S_n*,* $|\texttt{bridged}_n| \leq 2K \cdot |\texttt{succ}_n| \leq 2K^2$*, and every entry* $(p \mapsto b) \in \texttt{bridged}_n$ *is backed by a currently referenced successor* $b \in \text{Referenced}(S_n)$ *and refreshed on every stabilization or successor-sync round with* b *without wall-clock TTL timers.*

B. Gap-Aware `find_successor` Preserving `RoutesClockwise`

A critical invariant of `rings-core` (`invariants.rs`) is `RoutesClockwise`:

$$\text{find_successor}(S_n, id) = \text{Remote}(next, id) \implies next \in (n, id), \quad (14)$$

i.e., $0 < \text{d}(n, next) < \text{d}(n, id)$. Forwarding to $\text{head}(S_n)$ when $\text{d}(n, id) < \text{d}(n, \text{head}(S_n))$ would violate (14). However, when $id \in (n, \text{head}(S_n)]$ $(id \neq n)$, any peer $p \in (n, \text{head}(S_n))$ advertised by the bridge is *already present* in $\text{dom}(\texttt{bridged}_n)$! Furthermore, any direct predecessor $p \in \texttt{preds}_n$ is a known live member of the ring. We therefore define gap-aware `find_successor` as:

1) If $id \neq n$ and $id \in \texttt{preds}_n$, return Local(id).
2) Let $h = \text{head}(S_n)$ and $t = \text{d}(n, id)$. If $t \leq \text{d}(n, h)$:
   - If $id = n$, return Local(h).
   - Let $B_{\text{gap}} = \{p \in \text{dom}(\texttt{bridged}_n) \mid \texttt{bridged}_n(p) \in \text{Referenced}(S_n) \wedge t \leq \text{d}(n, p) < \text{d}(n, h)\}$.
   - If $B_{\text{gap}} \neq \emptyset$, return $\text{Local}\big(\arg\min_{p \in B_{\text{gap}}} \text{d}(n, p)\big)$; otherwise return Local$(h)$.
3) If $t > \text{d}(n, h)$, select $next = \arg\max\{\text{d}(n, p) \mid p \in \text{Referenced}(S_n) \setminus \{n\},\ \text{d}(n, p) < t\}$ (scanning $\texttt{fingers}_n$, $\texttt{succ}_n$, and $\texttt{preds}_n$), returning Remote$(next, id)$ if such a p exists and Local(h) otherwise.

Lemma 4 (Preservation of `RoutesClockwise`)**.** *Gap-aware* `find_successor` *satisfies* (14) *for every representable state* S_n *and every target* $id \in \mathbb{Z}_{2^{160}}$*.*

Proof. Clauses 1 and 2 return Local$(\cdot)$, which satisfies (14) vacuously. Clause 3 emits Remote$(next, id)$ only for candidates satisfying $next \neq n$ and $\text{d}(n, next) < t = \text{d}(n, id)$. □

V. Bridge-Delegated Delivery and Route Laws

Payload delivery in `rings-core` (`dht::delivery`) tags each frame with a stage $\sigma = (\texttt{via}, \texttt{handed_off}) \in \text{Option}\langle\text{Did}\rangle \times \mathbb{B}$, where $\text{aim}(\sigma, T) = \texttt{via}.\,\text{unwrap_or}(T)$. We extend delivery_step$(S_n, T, \sigma, \text{Linked}_n)$ with two passive bridge rules while keeping the wire format of RouteStage unchanged:

1) **Direct destination links:** If $\text{Linked}_n(T)$, return $\text{Some}\big((T, \sigma)\big)$.
2) **Rule B1 (1-Hop Bridge Target Delegation):** If $\sigma.\texttt{via} = \text{None}$ and $\texttt{bridged}_n(T) = \text{Some}(b)$ with $b \neq n \wedge \text{Linked}_n(b)$, return
$$\text{Some}\Big(\big(b,\ (\text{Some}(b), \sigma.\texttt{handed_off})\big)\Big). \quad (15)$$
3) **Explicit aim resolution:** Let $a = \text{aim}(\sigma, T)$. If $a \neq T$: if $a = n$ return None (the bridge lost its link to T); if $\text{Linked}_n(a)$ return $\text{Some}\big((a, \sigma)\big)$.

4) **Rule B2 (Gap-Bridge Handoff):** Evaluate route_toward$(S_n, a, \text{Linked}_n)$ over $\text{Known}(S_n) \cap \text{Linked}_n$, where $\text{Known}(S_n) = \text{Referenced}(S_n)$.
   - On Forward(p), return Some$\big((p, \sigma)\big)$.
   - On Handoff(h) when $\neg\sigma.\texttt{handed_off}$: if $\sigma.\texttt{via} =$ None and there exists $(p \mapsto b) \in \texttt{bridged}_n$ such that $b \neq n$, $\text{Linked}_n(b)$, and $\text{d}(n,p) < \text{d}(n,h)$ (i.e., b bridges into the local gap (n,h) in which a fell), select $b^* = b$ (minimizing $\text{d}(n,p)$) and return Some$\big((b^*, (\text{None}, \top))\big)$; otherwise return Some$\big((h, (\sigma.\texttt{via}, \top))\big)$.
   - Otherwise return None.

Theorem 1 (Preservation of Route Safety Laws)**.** *For any graph $G = (V, L)$, any collection of local views $\{S_n\}_{n \in V}$, any origin $n_0 \in V$, destination $T \in V$, and initial stage σ_0, the sequence of states $(n_0, \sigma_0), (n_1, \sigma_1), \ldots$ produced by* delivery_step *satisfies:*

1) ***Linked:*** *Every hop (n_k, n_{k+1}) satisfies $\{n_k, n_{k+1}\} \in L$.*
2) ***NeverOvershoots:*** *For every hop with $n_{k+1} \neq T$, $n_{k+1} \neq \text{aim}(\sigma_{k+1}, T)$, and $\neg(\sigma_{k+1}.\texttt{handed_off} \wedge \neg\sigma_k.\texttt{handed_off})$, $\text{d}(n_k, n_{k+1}) < \text{d}(n_k, \text{aim}(\sigma_{k+1}, T))$.*
3) ***Acyclic and Bounded:*** *No state (n_k, σ_k) with $n_k \neq T$ is visited twice, and the total number of hops is at most $2|V|$.*
4) ***Exact Local Delivery:*** *If $\sigma_0.\texttt{via} =$ None and the route terminates undelivered, no visited node n_k had $\{n_k, T\} \in L$.*

Proof. (1) Every branch returning Some$((p, \cdot))$ checks $\text{Linked}_{n_k}(p)$ directly. (2) Under Rule B1 (15), $n_{k+1} = b = \text{aim}(\sigma_{k+1}, T)$, so the precondition $n_{k+1} \neq \text{aim}(\sigma_{k+1}, T)$ is false. Under Rule B2, $\sigma_k.\texttt{handed_off} = \bot$ and $\sigma_{k+1}.\texttt{handed_off} = \top$, so the crossing guard holds. Under Forward(p), $p \in (n_k, a]$ with $p \neq a \implies \text{d}(n_k, p) < \text{d}(n_k, a)$. (3) Within each fixed stage σ, $\text{d}(n_k, \text{aim}(\sigma, T))$ strictly decreases in $\mathbb{N}$ on every Forward step, visiting at most $|V| - 1$ nodes. The boolean `handed_off` transitions $\bot \to \top$ at most once. Rule B1 transitions `via` from None to Some(b) and jumps directly to $n_{k+1} = b$; at b, either $\text{Linked}_b(T)$ delivers in one final hop ($n_{k+2} = T$) or $a = b =$ local terminates immediately with None. Hence Rule B1 cannot introduce any cycle and adds at most one intermediate node b (which was not the destination). Thus the route visits at most $2|V|$ hops. (4) Clause 1 checks $\text{Linked}_{n_k}(T)$ before any other rule, so no node linked to T can fail to deliver to T. □

Theorem 2 (All-Pairs Delivery Across Bridge-Connected Cliques)**.** *Let $G = (V, L)$ consist of two cliques V_1, V_2 sharing bridge node C with $|V_1 \setminus \{C\}| \leq K$ and $|V_2 \setminus \{C\}| \leq K$. Once each node in $V_1 \cup V_2$ has processed one stabilization report from its sub-clique head and (for nodes whose non-head successor is C) one successor-sync report from C, every ordered pair $(u, v) \in V \times V$ $(u \neq v)$ delivers in at most 2 hops (and in at most 3 hops even before non-head successor sync with C completes).*

TABLE I
MECHANIZED VERIFICATION OBLIGATIONS IN `rings-core`

Obligation / Law	Module	Coverage
(10) Empty-confirmed commit	`transport`	Unit + int.
Candidate dial backoff $\mathcal{B}_n$	`transport`	Unit test
Def. 3 Multi-predecessor	`topology`	Unit + prop.
Lemma 1 Retention	`topology`	Unit test
Lemma 2 Sub-clique proof	`topology`	Unit test
Zero-TTL `bridged` lifecycle	`topology`	Unit test
(14) `RoutesClockwise`	`topology`	Property test
Theorem 1 Route safety	`delivery`	Exhaustive
Theorem 2 Two-clique delivery	`delivery`	Model test
Stage 6 rejoin BFS invariants	`rejoin_model`	6 BFS models

Proof. If u and v belong to the same clique V_i, then $\{u, v\} \in L$ and Clause 1 delivers in 1 hop ($u \to v$). Otherwise, suppose $u \in V_i \setminus \{C\}$ and $v \in V_j \setminus \{C\}$ $(i \neq j)$. By Lemma 1, $\text{Referenced}(S_C) = (V_1 \cup V_2) \setminus \{C\}$, and every peer $v \in V_j \setminus \{C\}$ appears in C's `TopoInfo` ($\texttt{succ}_C \cup \texttt{preds}_C$). After u receives C's `TopoInfo` (via stabilization if $\text{head}(u) = C$, or via successor sync since $C \in \texttt{succ}_u$ when $|V_i \setminus \{C\}| \leq K$), $\texttt{bridged}_u(v) = \text{Some}(C)$. Since $\{u, C\} \in L$ and $\{C, v\} \in L$, Rule B1 at u forwards to C with stage $(\text{Some}(C), \bot)$, and Clause 1 at C delivers to v, yielding the 2-hop path $u \to C \to v$. Even before a non-head node $u \in V_i$ completes successor sync with C, one greedy or handoff hop inside V_i reaches the sub-clique predecessor $p_i = \text{Pred}_{V_i}(C)$ whose head is C; p_i holds $\texttt{bridged}_{p_i}(v) = \text{Some}(C)$ from stabilization and applies Rule B1 to deliver via $p_i \to C \to v$ in 3 hops. □

VI. Implementation and Verification Summary

Table I summarizes the mechanized obligations mapping each protocol law to the test suites in `crates/core`.

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